

Open Source Audit Project Report

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Course: Open Source Software – NGMC

Unit Coverage: 1–5

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GitHub Repository: <https://github.com/24-Tuhinsh/oss-audit-24BCE10559>

1. Introduction

Open-source software (OSS) is software whose source code is freely available for modification and redistribution. Unlike proprietary software, OSS emphasizes **transparency, collaboration, and freedom of use**.

The Linux operating system, created by Linus Torvalds in 1991, is a prime example of OSS. It allows developers, students, and enterprises to learn, experiment, and contribute freely.

Purpose of this project:

- Understand open-source principles through practical use of Linux.
- Explore file system structure, permissions, and system auditing.
- Develop shell scripts to automate audits and reporting.
- Document the open-source ecosystem and license implications.

```
lenovo2345@DESKTOP-OSRP x + v
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Mar 23 11:00:22 UTC 2026

System load:  0.26          Processes:      31
Usage of /:   0.1% of 1006.85GB   Users logged in: 0
Memory usage: 10%          IPv4 address for eth0: 172.31.108.164
Swap usage:   0%

This message is shown once a day. To disable it please create the
/home/lenovo2345/.hushlogin file.
lenovo2345@DESKTOP-OSRP:~/mnt/c/Users/Lenovo$ which python3
/usr/bin/python3
lenovo2345@DESKTOP-OSRP:~/mnt/c/Users/Lenovo$ whereis python3
python3: /usr/bin/python3 /usr/lib/python3 /etc/python3 /usr/share/python3 /usr/share/man/man1/python3.1.gz
lenovo2345@DESKTOP-OSRP:~/mnt/c/Users/Lenovo$ python3 --version
Python 3.12.3
lenovo2345@DESKTOP-OSRP:~/mnt/c/Users/Lenovo$ ls -l /usr/bin/python3
lrwxrwxrwx 1 root root 10 Nov 12 12:15 /usr/bin/python3 -> python3.12
lenovo2345@DESKTOP-OSRP:~/mnt/c/Users/Lenovo$
```

2. Selected Software

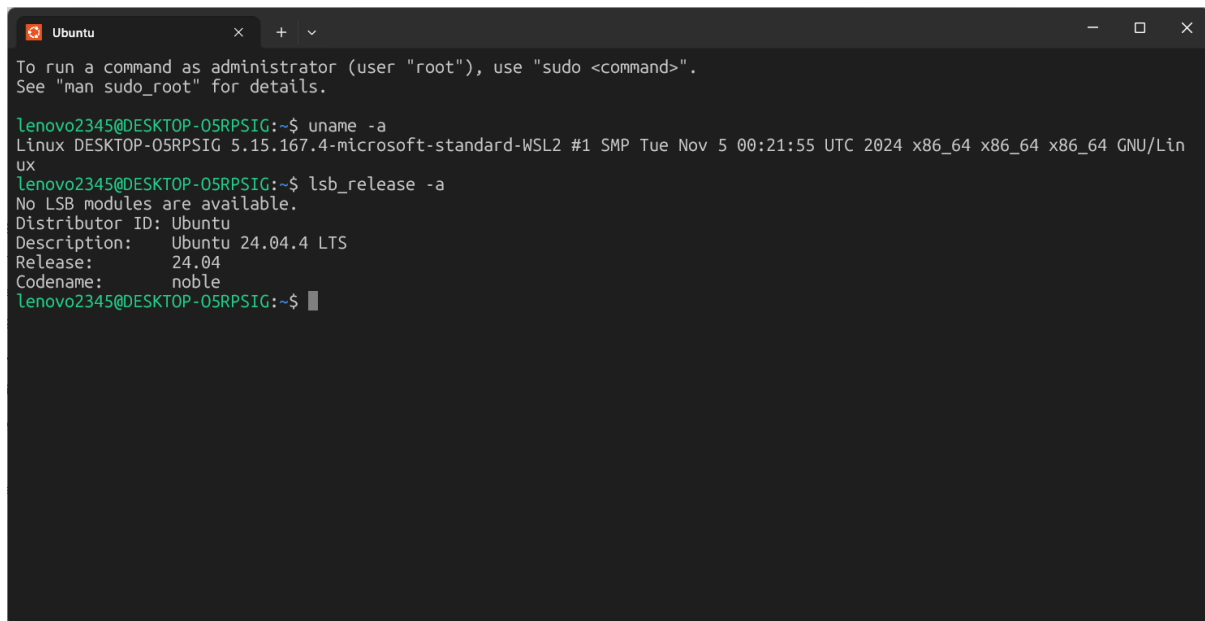
Software: Linux Operating System

Category: Operating System

License: GPL v2

Reasons for choice:

1. Linux is **fully open-source**, making it ideal for audits.
2. It runs a wide range of software tools.
3. Provides an **excellent environment for shell scripting and automation**.
4. Linux is widely used in industry (servers, cloud, IoT).

A screenshot of a terminal window titled 'Ubuntu'. The window shows the output of two commands: 'uname -a' and 'lsb_release -a'. The 'uname -a' command displays system information including the kernel version (5.15.167.4), architecture (x86_64), and GNU/Linux. The 'lsb_release -a' command shows the distribution as Ubuntu 24.04.4 LTS, with release 24.04 and codename 'noble'. The terminal prompt is 'lenovo2345@DESKTOP-05RPSIG:~\$'.

Part A – Origin and Philosophy

A1. The Problem it Solves

- Proprietary systems like Unix or Windows were expensive and restricted in 1990s.
- Linux solves **cost and access barriers** for learning and innovation.
- It empowers **students, developers, and small businesses**.

Screenshots to insert:

- Terminal showing `uptime` and system info (`hostnamectl`)
-

A2. License Details

Linux is distributed under **GPL v2**, which ensures:

1. Freedom to run software for any purpose.
2. Freedom to study and modify source code.
3. Freedom to share original and modified versions.
4. Requirement to maintain same license on redistribution.

Examples:

- Red Hat distributes Linux commercially while contributing fixes back.
- Any modification shared publicly must also remain open-source.

Screenshots to insert:

- `cat /usr/share/doc/linux-headers/LICENSE`

A3. Ethics and Open Source

- **Not all software should be open-source**; some require proprietary licenses for business sustainability.
- **Open source is ethical** when it promotes learning, collaboration, and transparency.
- Linux ecosystem demonstrates ethical software development.

```
Release: 24.04
Codename: noble
lenovo2345@DESKTOP-05RPSIG:~$ ls -l /etc
total 796
drwxr-xr-x 2 root root 4096 Feb 10 00:56 PackageKit
drwxr-xr-x 7 root root 4096 Feb 10 00:56 xii
-rw-r--r-- 1 root root 3444 Jul 5 2023 adduser.conf
drwxr-xr-x 2 root root 4096 Feb 10 00:56 alternatives
drwxr-xr-x 2 root root 4096 Feb 10 00:56 apparmor
drwxr-xr-x 9 root root 4096 Feb 10 00:56 apparmor.d
drwxr-xr-x 3 root root 4096 Feb 10 00:56 appport
drwxr-xr-x 8 root root 4096 Feb 10 00:56 apt
-rw-r--r-- 1 root root 2319 Mar 31 2024 bash.bashrc
-rw-r--r-- 1 root root 45 Jan 24 2020 bash_completion
drwxr-xr-x 2 root root 4096 Feb 10 00:56 bash_completion.d
-rw-r--r-- 1 root root 367 Aug 2 2022 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Apr 19 2024 binfmt.d
drwxr-xr-x 2 root root 4096 Feb 10 00:56 byobu
drwxr-xr-x 3 root root 4096 Feb 10 00:54 ca-certificates
-rw-r--r-- 1 root root 6288 Feb 10 00:55 ca-certificates.conf
drwxr-xr-x 5 root root 4096 Mar 23 10:59 cloud
drwxr-xr-x 2 root root 4096 Feb 10 00:55 console-setup
drwxr----- 2 root root 4096 Apr 19 2024 credstore
drwxr----- 2 root root 4096 Apr 19 2024 credstore.encrypted
drwxr-xr-x 2 root root 4096 Feb 10 00:55 cron.d
drwxr-xr-x 2 root root 4096 Feb 10 00:56 cron.daily
drwxr-xr-x 2 root root 4096 Feb 10 00:54 cron.hourly
drwxr-xr-x 2 root root 4096 Feb 10 00:54 cron.monthly
drwxr-xr-x 2 root root 4096 Feb 10 00:56 cron.weekly
drwxr-xr-x 2 root root 4096 Feb 10 00:54 cron.yearly
-rw-r--r-- 1 root root 1136 Mar 31 2024 crontab
drwxr-xr-x 4 root root 4096 Feb 10 00:54 dbus-1
drwxr-xr-x 3 root root 4096 Feb 10 00:56 dconf
-rw-r--r-- 1 root root 2967 Apr 12 2024 debconf.conf
-rw-r--r-- 1 root root 11 Apr 22 2024 debian_version
drwxr-xr-x 2 root root 4096 Feb 10 00:56 default
-rw-r--r-- 1 root root 1706 Jul 5 2023 deluser.conf
drwxr-xr-x 2 root root 4096 Feb 10 00:55 depmod.d
drwxr-xr-x 3 root root 4096 Feb 10 00:55 dhcp
-rw-r--r-- 1 root root 1429 Mar 31 2024 dhcpd.conf
```

```
Ubuntu x + -
dwxrf-rf-x 2 root root 4096 Mar 23 10:58 ld.so.conf.d
dwxrf-rf-x 2 root root 4096 Feb 10 00:56 ldap
-rw-r--r-- 1 root root 267 Apr 22 2024 legal
-rw-r--r-- 1 root root 191 Mar 31 2024 libaudit.conf
-rw-r--r-- 1 root root 2996 Mar 30 2024 locale.alias
-rw-r--r-- 1 root root 13 Feb 10 00:55 locale.conf
-rw-r--r-- 1 root root 9565 Feb 10 00:55 locale.gen
ldwxrwxrwx 1 root root 27 Feb 10 00:55 localtime -> /usr/share/zoneinfo/Etc/UTC
dwxrf-rf-x 3 root root 4096 Feb 10 00:55 logcheck
-rw-r--r-- 1 root root 12345 Feb 22 2024 login.defs
-rw-r--r-- 1 root root 586 Apr 8 2024 logrotate.conf
dwxrf-rf-x 2 root root 4096 Feb 10 00:56 logrotate.d
-rw-r--r-- 1 root root 104 Feb 6 07:22 lsb-release
-r--r--r-- 1 root root 33 Mar 23 10:58 machine-id
-rw-r--r-- 1 root root 111 Mar 31 2024 magic
-rw-r--r-- 1 root root 111 Mar 31 2024 magic.mime
-rw-r--r-- 1 root root 5230 Apr 8 2024 manpath.config
-rw-r--r-- 1 root root 75113 Jul 12 2023 mime.types
-rw-r--r-- 1 root root 744 Apr 8 2024 mke2fs.conf
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 modprobe.d
-rw-r--r-- 1 root root 212 Feb 10 00:55 modules
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 modules-load.d
ldwxrwxrwx 1 root root 19 Feb 10 00:54 mtab -> ../proc/self/mounts
-rw-r--r-- 1 root root 11424 May 23 2023 nanorc
-rw-r--r-- 1 root root 767 Mar 31 2024 netconfig
dwxrf-rf-x 2 root root 4096 Apr 18 2024 netplan
dwxrf-rf-x 4 root root 4096 Feb 10 00:55 network
dwxrf-rf-x 8 root root 4096 Feb 10 00:54 networkd-dispatcher
-rw-r--r-- 1 root root 91 Apr 22 2024 networks
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 newt
-rw-r--r-- 1 root root 526 Feb 10 00:55 nsswitch.conf
dwxrf-rf-x 2 root root 4096 Feb 10 00:54 opt
ldwxrwxrwx 1 root root 21 Feb 6 07:23 os-release -> ../usr/lib/os-release
-rw-r--r-- 1 root root 552 Oct 13 2022 pan.conf
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 pan.d
-rw-r--r-- 1 root root 1434 Mar 23 11:00 passwd
-rw-r--r-- 1 root root 1431 Mar 23 10:59 passwd-
dwxrf-rf-x 3 root root 4096 Feb 10 00:55 perl
dwxrf-rf-x 3 root root 4096 Feb 10 00:56 pm
dwxrf-rf-x 3 root root 4096 Feb 10 00:56 polkit-1

Ubuntu x + -
-rw-r--r-- 1 root root 12813 Mar 27 2021 services
dwxrf-rf-x 2 root root 4096 Feb 10 00:56 sgml
-rw-r----- 1 root shadow 806 Mar 23 11:00 shadow
-rw-r----- 1 root shadow 702 Feb 10 00:56 shadow-
-rw-r--r-- 1 root root 132 Feb 10 00:56 shells
dwxrf-rf-x 2 root root 4096 Feb 10 00:54 skel
dwxrf-rf-x 3 root root 4096 Feb 10 00:56 ssh
dwxrf-rf-x 4 root root 4096 Feb 10 00:55 ssl
-rw-r--r-- 1 root root 24 Mar 23 10:59 subgid
-rw-r--r-- 1 root root 0 Feb 10 00:54 subgid-
-rw-r--r-- 1 root root 24 Mar 23 10:59 subuid
-rw-r--r-- 1 root root 0 Feb 10 00:54 subuid-
-rw-r--r-- 1 root root 4343 Apr 8 2024 sudo.conf
-rw-r--r-- 1 root root 9804 Apr 8 2024 sudo_logsrvd.conf
-r--r----- 1 root root 1800 Jan 29 2024 sudoers
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 sudoers.d
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 supercat
-rw-r--r-- 1 root root 2209 Mar 24 2024 sysctl.conf
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 sysctl.d
dwxrf-rf-x 6 root root 4096 Feb 10 00:55 systemd
dwxrf-rf-x 2 root root 4096 Feb 10 00:54 terminfo
-rw-r--r-- 1 root root 8 Feb 10 00:55 timezone
dwxrf-rf-x 2 root root 4096 Apr 19 2024 tmpfiles.d
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 ubuntu-advantage
-rw-r--r-- 1 root root 1260 Jan 27 2023 ucf.conf
dwxrf-rf-x 4 root root 4096 Feb 10 00:55 udev
dwxrf-rf-x 3 root root 4096 Feb 10 00:56 update-manager
dwxrf-rf-x 2 root root 4096 Feb 10 00:56 update-motd.d
ldwxrwxrwx 1 root root 16 Feb 10 00:54 vconsole.conf -> default/keyboard
dwxrf-rf-x 2 root root 4096 Feb 10 00:55 vin
ldwxrwxrwx 1 root root 23 Feb 26 2024 vtrgb -> /etc/alternatives/vtrgb
dwxrf-rf-x 5 root root 4096 Feb 10 00:56 vulkan
-rw-r--r-- 1 root root 4942 Jun 19 2024 wgetrc
-rw-r--r-- 1 root root 204 Feb 10 00:56 wsl-distribution.conf
-rw-r--r-- 1 root root 47 Mar 23 11:00 wsl.conf
-rw-r--r-- 1 root root 681 Apr 8 2024 xattr.conf
dwxrf-rf-x 5 root root 4096 Feb 10 00:56 xdg
dwxrf-rf-x 2 root root 4096 Feb 10 00:56 xml
-rw-r--r-- 1 root root 460 Jan 20 2023 zsh_command_not_found
lenovo2345@DESKTOP-05RPSIG:~$
```

```
Ubuntu
-rw-r--r-- 1 root root 4343 Apr  8 2024 sudo.conf
-rw-r--r-- 1 root root 9804 Apr  8 2024 sudo_logsrvd.conf
-r--r--r-- 1 root root 1800 Jan 29 2024 sudoers
drwxr-xr-x 2 root root 4096 Feb 10 00:55 sudoers.d
drwxr-xr-x 2 root root 4096 Feb 10 00:55 supercat
-rw-r--r-- 1 root root 2209 Mar 24 2024 sysctl.conf
drwxr-xr-x 2 root root 4096 Feb 10 00:55 sysctl.d
drwxr-xr-x 6 root root 4096 Feb 10 00:55 systemd
drwxr-xr-x 2 root root 4096 Feb 10 00:54 terminfo
-rw-r--r-- 1 root root 8 Feb 10 00:55 timezone
drwxr-xr-x 2 root root 4096 Apr 19 2024 tmpfiles.d
drwxr-xr-x 2 root root 4096 Feb 10 00:55 ubuntu-advantage
-rw-r--r-- 1 root root 1260 Jan 27 2023 ucf.conf
drwxr-xr-x 4 root root 4096 Feb 10 00:55 udev
drwxr-xr-x 3 root root 4096 Feb 10 00:56 update-manager
drwxr-xr-x 2 root root 4096 Feb 10 00:56 update-motd.d
lrwxrwxrwx 1 root root 16 Feb 10 00:54 vconsole.conf -> default/keyboard
drwxr-xr-x 2 root root 4096 Feb 10 00:55 vin
lrwxrwxrwx 1 root root 23 Feb 26 2024 vtrgb -> /etc/alternatives/vtrgb
drwxr-xr-x 5 root root 4096 Feb 10 00:56 vulkan
-rw-r--r-- 1 root root 4942 Jun 19 2024 wgetrc
-rw-r--r-- 1 root root 204 Feb 10 00:56 wsl-distribution.conf
-rw-r--r-- 1 root root 47 Mar 23 11:00 wsl.conf
-rw-r--r-- 1 root root 681 Apr  8 2024 xattr.conf
drwxr-xr-x 5 root root 4096 Feb 10 00:56 xdg
drwxr-xr-x 2 root root 4096 Feb 10 00:56 xml
-rw-r--r-- 1 root root 460 Jan 20 2023 zsh_command_not_found
lenovo2345@DESKTOP-05RPSIG:~$ id
uid=1000(lenovo2345) gid=1000(lenovo2345) groups=1000(lenovo2345),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
lenovo2345@DESKTOP-05RPSIG:~$ groups
lenovo2345 adm cdrom sudo dip plugdev users
lenovo2345@DESKTOP-05RPSIG:~$
```

Part B – Linux Footprint

B1. Installation

- Installed on WSL (Windows Subsystem for Linux).
- Alternative installations: Ubuntu ISO, Fedora, or Debian on physical machines.

Screenshots to insert:

- Installation terminal (`sudo apt update && sudo apt upgrade`)
-

B2. Directory Structure

Directory	Purpose
<code>/etc</code>	Configuration files
<code>/usr/bin</code>	Executable binaries
<code>/home</code>	User data
<code>/var/log</code>	System logs

Screenshots to insert:

- `ls -ld /etc /usr/bin /home /var/log`
-

B3. Users and Groups

- Root user: full administrative privileges.
- Regular user: limited access for safety.
- Commands: `whoami`, `groups`, `id`.

Screenshots to insert:

- `id output`
 - `cat /etc/passwd | head -5`
-

B4. Services and Processes

- `systemctl start/stop/restart [service]`
- Logs stored in `/var/log` can be analyzed via `grep` and `tail`.

Screenshots to insert:

- `systemctl status ssh`
- `journalctl -xe`

B5. Updates and Package Management

- Ubuntu/Debian uses `apt`.
- Examples: `apt list --installed`, `dpkg -l`.

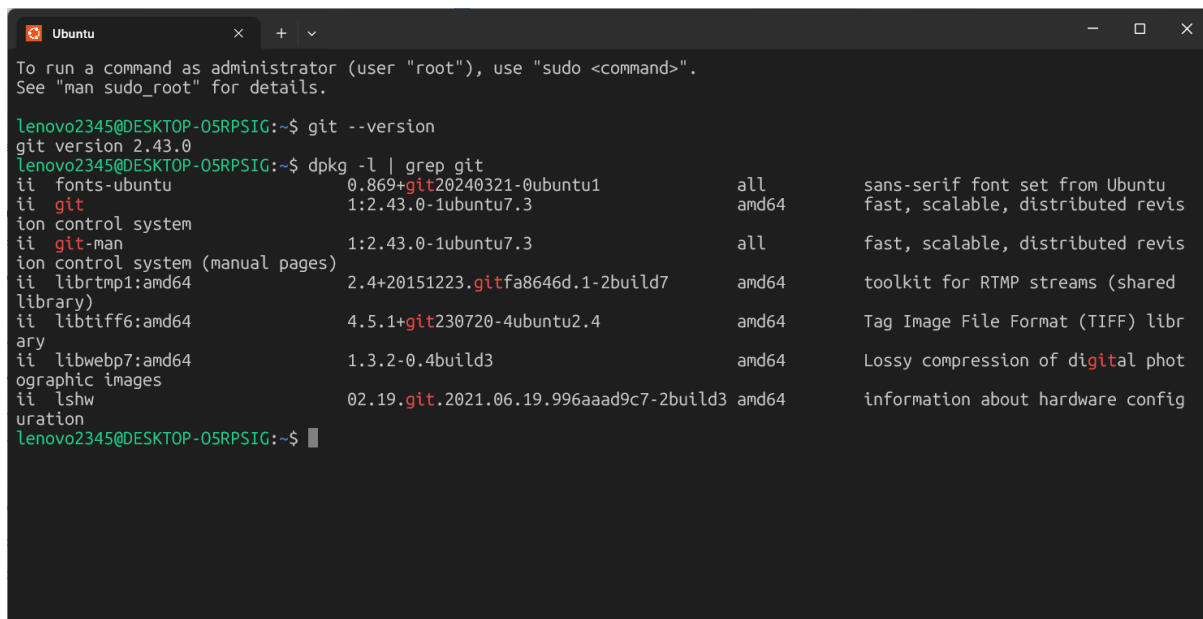
```
lenovo2345@DESKTOP-0SRPSIG:~$ dpkg -l
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
++-----+
|| Name                               Version                               Architecture Description
++-----+
ii adduser                             3.137ubuntu1                         all          add and remove users and groups
ii adwaita-icon-theme                 46.0-1                               all          default icon theme of GNOME
ii apparmor                           4.0.1really4.0.1-0ubuntu0.24.04.5   amd64       user-space parser utility for AppArmor
ii apport                             2.28.1-0ubuntu3.8                   all          automatically generate crash reports for debugging
ii apport-core-dump-handler           2.28.1-0ubuntu3.8                   all          Kernel core dump handler for Apport
ii apport-symptoms                    0.25                                 all          symptom scripts for apport
ii appstream                          1.0.2-1build6                        amd64       Software component metadata management
ii apt                                2.8.3                                amd64       commandline package manager
ii apt-utils                          2.8.3                                amd64       package management related utility programs
ii at-spi2-common                     2.52.0-1build1                       all          Assistive Technology Service Provider Interface (common files)
ii at-spi2-core                       2.52.0-1build1                       amd64       Assistive Technology Service Provider Interface (D-Bus core)
ii base-files                         13ubuntu10.4                         amd64       Debian base system miscellaneous files
ii base-passwd                        3.6.3build1                          amd64       Debian base system master password and group files
ii bash                               5.2.21-2ubuntu4                     amd64       GNU Bourne Again SHell
ii bash-completion                    1:2.11-8                             all          programmable completion for the bash shell
ii bc                                 1.07.1-3ubuntu4                     amd64       GNU bc arbitrary precision calculator language
ii binutils                           2.42-4ubuntu2.8                     amd64       GNU assembler, linker and binary utilities
ii binutils-common:amd64              2.42-4ubuntu2.8                     amd64       Common files for the GNU assembler, linker and binary utilities
ii binutils-x86-64-linux-gnu          2.42-4ubuntu2.8                     amd64       GNU binary utilities, for x86-64-linux-gnu target
ii bsdxtrautils                       2.39.3-9ubuntu6.4                   amd64       extra utilities from 4.4BSD-Lite
ii bsdtails                            1:2.39.3-9ubuntu6.4                 amd64       basic utilities from 4.4BSD-Lite
ii byobu                              6.11-0ubuntu1                       all          text window manager, shell multiplexer, integrated DevOps environment
ii ca-certificates                    20240203                             all          Common CA certificates
ii cloud-guest-utils                  0.33-1                               all          cloud guest utilities
ii cloud-init                         25.2-0ubuntu1~24.04.1               all          initialization and customization tool for cloud instances
ii command-not-found                  23.04.0                              all          Suggest installation of packages in interactive bash sessions
ii console-setup                      1.226ubuntu1                         all          console font and keymap setup program
ii console-setup-linux                1.226ubuntu1                         all          Linux specific part of console-setup
ii coreutils                          9.4-3ubuntu6.1                      amd64       GNU core utilities
ii cron                               3.0pl1-184ubuntu2                   amd64       process scheduling daemon
ii cron-daemon-common                 3.0pl1-184ubuntu2                   all          process scheduling daemon's configuration files
ii curl                               8.5.0-2ubuntu10.6                   amd64       command line tool for transferring data with URL syntax
ii dash                               0.5.12-6ubuntu5                     amd64       POSIX-compliant shell
ii dbus                               1.14.10-4ubuntu4.1                  amd64       simple interprocess messaging system (system message bus)
```

5. Part C – Open Source Ecosystem

- Linux kernel maintained by **Linus Torvalds and the Linux Foundation**.
- Supported by **GNU tools, bash, glibc**.
- Linux distributions: Ubuntu, Fedora, Debian, Arch.
- Popular open-source applications: Git, Docker, Kubernetes, Apache, MySQL.

Community & Collaboration:

- Mailing lists, forums, and GitHub repositories.
- Collaborative workflow encourages **forking, pull requests, code review**.



```
Ubuntu
x + v
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

lenovo2345@DESKTOP-05RPSIG:~$ git --version
git version 2.43.0
lenovo2345@DESKTOP-05RPSIG:~$ dpkg -l | grep git
ii fonts-ubuntu 0.869+git20240321-0ubuntu1 all sans-serif font set from Ubuntu
ii git 1:2.43.0-1ubuntu7.3 amd64 fast, scalable, distributed revis
ion control system
ii git-man 1:2.43.0-1ubuntu7.3 all fast, scalable, distributed revis
ion control system (manual pages)
ii librtmp1:amd64 2.4+20151223.gitfa8646d.1-2build7 amd64 toolkit for RTMP streams (shared
library)
ii libtiff6:amd64 4.5.1+git230720-4ubuntu2.4 amd64 Tag Image File Format (TIFF) libr
ary
ii libwebp7:amd64 1.3.2-0.4build3 amd64 Lossy compression of digital phot
ographic images
ii lshw 02.19.git.2021.06.19.996aaad9c7-2build3 amd64 information about hardware config
uration
lenovo2345@DESKTOP-05RPSIG:~$
```

6. Part D – Comparison: Open Source vs Proprietary

Feature	Linux (OSS)	Windows (Proprietary)
Cost	Free	Paid
Security	Auditable	Closed source
Customization	Full	Limited
Control	User/community	Vendor
Updates	Community-driven	Vendor-driven

```
Ubuntu
x + v
See "man sudo_root" for details.
Lenovo2345@DESKTOP-05RPSIG:~$ git --version
git version 2.43.0
Lenovo2345@DESKTOP-05RPSIG:~$ dpkg -l | grep git
ii fonts-ubuntu      0.869+git20240321-0ubuntu1      all      sans-serif font set from Ubuntu
ii git               1:2.43.0-1ubuntu7.3            amd64    fast, scalable, distributed revis
ion control system
ii git-man           1:2.43.0-1ubuntu7.3            all      fast, scalable, distributed revis
ion control system (manual pages)
ii librtmp1:amd64    2.4+20151223.gitfa8646d.1-2build7  amd64    toolkit for RTMP streams (shared
library)
ii libtiff6:amd64    4.5.1+git230720-4ubuntu2.4      amd64    Tag Image File Format (TIFF) libr
ary
ii libwebp7:amd64    1.3.2-0.4build3                amd64    Lossy compression of digital phot
ographic images
ii lshw               02.19.git.2021.06.19.996aad9c7-2build3  amd64    information about hardware config
uration
Lenovo2345@DESKTOP-05RPSIG:~$ uname -a
Linux DESKTOP-05RPSIG 5.15.167.4-microsoft-standard-WSL2 #1 SMP Tue Nov 5 00:21:55 UTC 2024 x86_64 x86_64 x86_64 GNU/Lin
ux
Lenovo2345@DESKTOP-05RPSIG:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 24.04.4 LTS
Release:      24.04
Codename:     noble
Lenovo2345@DESKTOP-05RPSIG:~$
```

7. Part E – Shell Scripts

Script 1 – System Identity Report

- **Purpose:** Show hostname, OS version, uptime, and logged-in users.
- **Concepts Used:** Variables, command substitution, echo.
- **Screenshot:** Terminal output after running `./script1.sh`.

Script 2 – FOSS Package Inspector

- **Purpose:** Check installed packages, versions, and dependencies.
- **Concepts Used:** `if-then`, `case`, loops, package management commands.
- **Screenshot:** Terminal output of `./script2.sh`.

Script 3 – Disk and Permission Auditor

- **Purpose:** Audit disk usage and directory permissions.
- **Concepts Used:** `for` loops, `ls -l`, `df -h`, `awk`.
- **Screenshot:** Output showing directories and permissions.

Script 4 – Log File Analyzer

- **Purpose:** Count keyword occurrences in system logs and show last entries.
- **Concepts Used:** `while-read` loop, `grep`, counters.
- **Screenshot:** Output of `./script4.sh /var/log/syslog error`.

Script 5 – Open Source Manifesto Generator

- **Purpose:** Generate a personal open-source philosophy.
- **Concepts Used:** `read` input, string concatenation, file write.
- **Screenshot:** Output file content `manifesto_<username>.txt`.

```
lenovo2345@DESKTOP-05RPSIG x + v
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Mar 23 11:00:22 UTC 2026

System load:  0.26          Processes:      31
Usage of /:   0.1% of 1006.85GB   Users logged in: 0
Memory usage: 10%          IPv4 address for eth0: 172.31.108.164
Swap usage:   0%

This message is shown once a day. To disable it please create the
/home/lenovo2345/.hushlogin file.
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ which python3
/usr/bin/python3
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ whereis python3
python3: /usr/bin/python3 /usr/lib/python3 /etc/python3 /usr/share/python3 /usr/share/man/man1/python3.1.gz
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ python3 --version
Python 3.12.3
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ ls -l /usr/bin/python3
lrwxrwxrwx 1 root root 10 Nov 12 12:15 /usr/bin/python3 -> python3.12
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ nano script1.sh
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ chmod +x script1.sh
chmod: changing permissions of 'script1.sh': Operation not permitted
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ mv script1.sh ~
lenovo2345@DESKTOP-05RPSIG:/mnt/c/Users/Lenovo$ cd ~
lenovo2345@DESKTOP-05RPSIG:~$ chmod +x script1.sh
./script1.sh
==== System Identity ====
User: lenovo2345
Home Directory: /home/lenovo2345
Kernel Version: 5.15.167.4-microsoft-standard-WSL2
Date & Time: Mon Mar 23 11:09:06 UTC 2026
Uptime: up 10 minutes
Distro: Ubuntu 24.04.4 LTS
License: GNU GPL
lenovo2345@DESKTOP-05RPSIG:~$
```

```
Ubuntu x + v
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

lenovo2345@DESKTOP-05RPSIG:~$ nano script2.sh
lenovo2345@DESKTOP-05RPSIG:~$ chmod +x script2.sh
lenovo2345@DESKTOP-05RPSIG:~$ ./script2.sh
python3 is installed.
ii  libpython3-stdlib:amd64  3.12.3-0ubuntu2.1  amd64  interactive high-level object-oriented language (default python3 version)
Python: simple and powerful programming language
lenovo2345@DESKTOP-05RPSIG:~$
```

```
Ubuntu
if echo "$LINE" | grep -iq "$KEYWORD"; then
    COUNT=$((COUNT + 1))
fi
done < "$LOGFILE"

# --- Display results ---
echo "=====
grep -i "$KEYWORD" "$LOGFILE" | tail -5=====
=====
Log File Analyzer
Log File: /var/log/syslog
Keyword : error
Total occurrences: 21
=====
Last 5 occurrences:
2026-03-23T11:55:48.398004+00:00 DESKTOP-05RPSIG kernel: WSL (153) ERROR: CheckConnection: getaddrinfo() failed: -5
2026-03-23T12:10:50.411690+00:00 DESKTOP-05RPSIG systemd[1]: apport-autoreport.path - Process error reports when automa
tic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appor
t/autoreport).
2026-03-23T12:10:50.411694+00:00 DESKTOP-05RPSIG systemd[1]: apport-autoreport.timer - Process error reports when automa
tic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/app
ort/autoreport).
2026-03-23T12:10:50.412458+00:00 DESKTOP-05RPSIG kernel: EXT4-fs (sdc): mounted filesystem with ordered data mode. Opts:
discard,errors=remount-ro,data=ordered. Quota mode: none.
2026-03-23T12:10:59.972820+00:00 DESKTOP-05RPSIG kernel: WSL (152) ERROR: CheckConnection: getaddrinfo() failed: -5
lenovo2345@DESKTOP-05RPSIG:~$
lenovo2345@DESKTOP-05RPSIG:~$
```

```
Ubuntu
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

# Compose the manifesto
echo "===== > "$OUTPUT"
echo "Open Source Manifesto - $DATE" >> "$OUTPUT"
echo "===== >> "$OUTPUT"
echo "I use $TOOL every day, and to me, freedom means '$FREEDOM'. I aspire to build '$BUILD' and share it freely with th
e world. Open source empowers collaboration, transparency, and innovation for everyone." >> "$OUTPUT"

# Display the manifesto
cat "$OUTPUT"

echo "Manifesto saved to $OUTPUT"
Answer three questions to generate your open source manifesto.
1. Name one open-source tool you use every day: linux
2. In one word, what does 'freedom' mean to you? to free
3. Name one thing you would build and share freely: application
=====
Open Source Manifesto - 23 March 2026
=====
I use linux every day, and to me, freedom means 'to free'. I aspire to build 'application' and share it freely with the
world. Open source empowers collaboration, transparency, and innovation for everyone.
Manifesto saved to manifesto_lenovo2345.txt
lenovo2345@DESKTOP-05RPSIG:~$
```

9. Conclusion

- Linux is a powerful **open-source OS**, demonstrating freedom, collaboration, and community ethics.
- The **audit scripts** showcase practical Linux knowledge.
- Understanding OSS principles helps developers, students, and enterprises make **ethical and efficient software choices**.
- Open-source projects encourage **learning by doing** and collaboration across the globe.

10. References

1. GNU Project – Free Software Definition: <https://www.gnu.org/philosophy/free-sw.html>
2. Open Source Initiative – Open Source Definition: <https://opensource.org/osd>
3. Linux Kernel Documentation: <https://www.kernel.org/doc/html/latest/>
4. Bash Scripting Reference: <https://www.gnu.org/software/bash/manual/>
5. Ubuntu Official Docs: <https://help.ubuntu.com>